

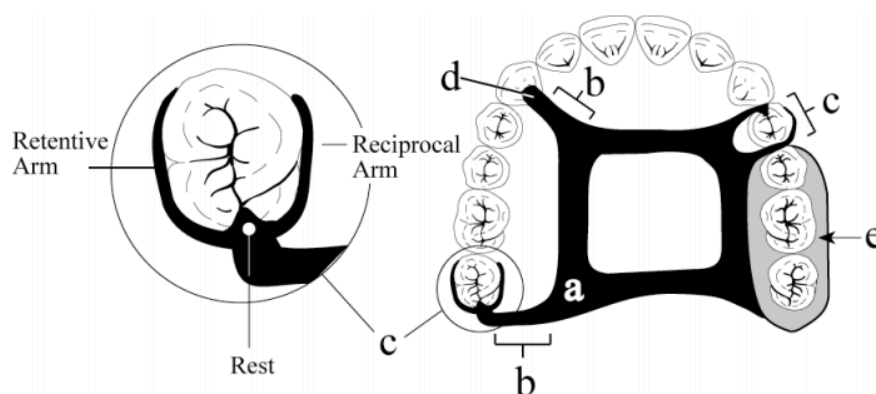
Components of Definitive Removable Partial Denture

Learning objectives

- Identify the components of RPD
 - Define major connectors of RPD
 - Distinguish different types of maxillary RPD
-

All definitive removable partial dentures have the following components:

- Major Connector:** The unit of a removable partial denture that connects the parts of one side of the dental arch to those of the other side.
- Minor Connector:** A unit of a partial denture that connects other components (i.e. direct retainer, indirect retainer, denture base, etc.) to the major connector.
- Direct Retainer:** A unit of a partial denture that provides retention against dislodging forces. A direct retainer is commonly called a 'clasp' or 'clasp unit' and is composed of four elements: a rest, a retentive arm, a reciprocal arm and a minor connector.
- Indirect Retainer:** A unit of a Class I or II partial denture that prevents or resists movement or rotation of the base(s) away from the residual ridge. The indirect retainer is usually composed of one component, a rest.
- Denture Base:** The unit of a partial denture that covers the residual ridges and supports the denture teeth.



Major Connectors

When using a removable partial denture, the prosthesis must extend to both sides of the arch. This enables the direction of functional forces to support teeth and tissue for optimum stability.

Functions of a major connector include

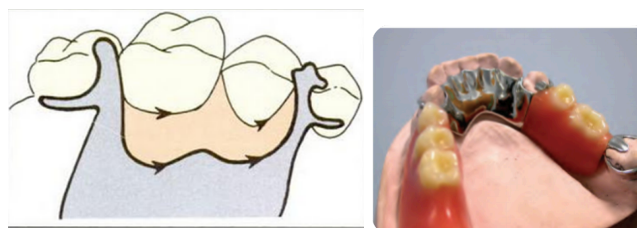
1. Unification of the major parts of the prosthesis.
2. Distribution of the applied force throughout the arch to selected teeth and tissue.
3. Cross-arch stabilization

Requirements of a major connector design

1. Rigidity: a major connector can be made more rigid by:
 - a. Using more rigid alloys (Chrome-cobalt > Gold alloys)
 - b. Using half-round or half-pear-shaped bars, which are more rigid than flat bars.
 - c. Increasing the bulk as the length increases.
2. No interference with soft tissue.
3. Minimise food impaction
4. Unobtrusive: smooth transitions, all line angles and edges should be smooth and rounded, bulk should be reduced enough to avoid interference with speech or appearance, yet thick enough to ensure rigidity.

Borders of major connectors should:

1. Run parallel to the gingival margin of the remaining teeth in both arches.



2. Be placed at least 3-4mm away from the free gingival margins in the mandibular arch and at least 6 mm in the maxillary arch. Otherwise, the major connectors should cover the lingual surfaces of the teeth as a plate.
3. Not end on the crest of rugae or at the free gingival margin.
4. Cross the maxillary midline at right angles, not diagonally. It has been suggested that the tongue will accept symmetrically placed components far more readily than those placed without regard for symmetry.

Maxillary Major Connectors

- A. Anterior-posterior palatal strap (bar)
- B. Full palatal plate
- C. Single palatal strap and palatal bar
- D. U-shaped palatal connector

A. Anterior-Posterior Palatal Strap (Bar)

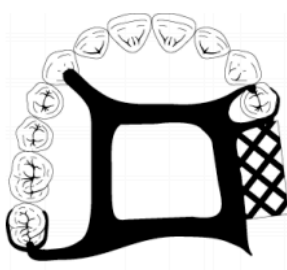
The anterior-posterior palatal strap provides maximum rigidity and minimum bulk. The strength of this connector lies in the fact that the anterior and posterior sections are joined together by longitudinal connectors on either side. Each component braces the other against possible torque and flexure.

It can be used in most maxillary partial denture designs and is especially useful in cases with a torus palatinus.

A variation of the double palatal strap is the anterior-posterior palatal bar connector, where the palatal connector elements are narrower antero-posteriorly (less than 8mm). Due to the narrowness of the elements, the bars must have greater bulk for rigidity, and thus the design is more objectionable to the patient.

Contraindications

1. Where the palatal opening is less than 15mm anteroposteriorly or mediolaterally.
2. Where simpler major connectors can be used.



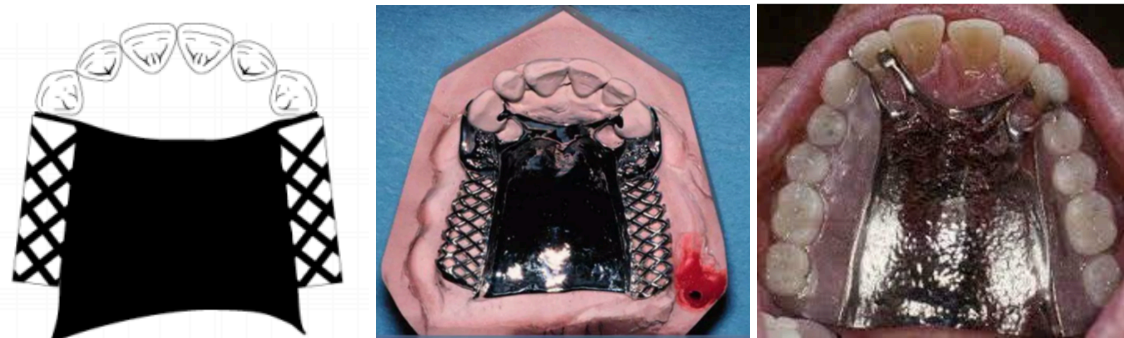
B. Full Palatal Plate

The full palatal plate is indicated when maximum tissue support is required. It should be the major connector of choice in long distal extension cases or where six or fewer anterior teeth remain.

The full palatal plate is usually not used in the presence of a torus palatinus. The full palatal connector should be fabricated of a uniformly thin metal plate with an accurate anatomic reproduction of the rugae configuration (improves strength and rigidity).

It should cover the same area as a complete denture posteriorly. The large surface area of contact with the mucosa enhances the potential for retention. This may be accomplished in one of two ways.

1. One method is to use a complete cast plate that extends to the junction of the hard and soft palates.
2. The other method is to use a cast major connector anteriorly with retentive means posteriorly, for the attachment of an acrylic-resin denture base that extends posteriorly to the posterior palatal seal area.



C. Single Palatal strap and palatal bar

This type of connector can be 8mm wide (strap) or narrower (bar), depending upon its location or the need for strength or support of the denture.

Indications:

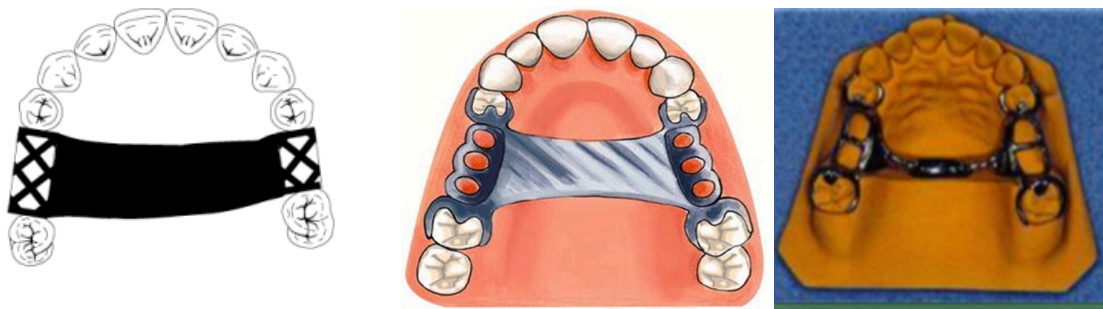
- They should only be used in tooth-supported cases (Class III with/without modifications) where no other connector can be used.

Contraindications:

1. It should never be used in cases involving distal extensions or replacement of anterior teeth.
2. A large torus
3. has many teeth that are being replaced.

A single palatal strap (bar) is better NOT to be placed anterior to the second premolar position; otherwise, its bulk may produce noticeable discomfort and alteration of speech.

The palatal strap is similar to the palatal bar, but with a broader area of contact, providing better stabilisation and stress-distributing properties with minimum bulk. Therefore palatal strap is preferable to the palatal bar for posterior tooth-supported cases. A single palatal bar is usually objectionable due to its bulk.



D. **U-shaped palatal connector** (anterior palatal strap, horseshoe, open ring, open doughnut)

From a biomechanical standpoint, the palatal horseshoe is a poor connector and should never be used unless necessary.

To be rigid, the horseshoe connector must have bulk in the rugae area where the tongue requires freedom. Without sufficient bulk, the U-shaped design becomes flexible and allows movement posteriorly.

In distal extension partial dentures, where posterior tooth support is non-existent, movement is particularly noticeable and is traumatic to the residual ridge, causing continual irritation during function. The anterior palatal plate can be prescribed in the following cases

1. where the torus palatinus prohibits the use of other connectors,
2. in class IV,
3. and in class III (Mod 1), with an anterior edentulous space where cross-arch force distribution is not important.

